Complexity of Nonlinear Conjugate Gradient Methods on Strict Saddle Functions: The Line-Search Setting

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May 4, 2026

Abstract

We study the complexity of nonlinear conjugate gradient (NCG) methods on strict saddle functions with backtracking line-search. We consider a hybrid algorithm that combines an NCG step with a modified restart condition and a negative-curvature direction step, and we establish complexity bounds in three regimes corresponding to the strict saddle decomposition: large-gradient (Regime 1), negative-curvature (Regime 2), and strong-convexity neighborhoods of local minimizers (Regime 3). The decrease lemmas in the NCG phase are inherited from [1]; the negative-curvature phase analysis is independent of the line search and is reused from [3]; and the strong-convexity phase is analyzed under a phase split based on the magnitude of the gradient.

Contents

1 Introduction

[To be written. Motivation, related work, contributions, organization.]

2 Setting and Algorithm

2.1 Problem setting

We consider the unconstrained minimization problem

$$\min_{x \in \mathbb{R}^d} f(x), \tag{1}$$

where $f\colon \mathbb{R}^d \to \mathbb{R}$ is nonconvex and satisfies the strict saddle property. We make the following standing assumptions.

Assumption 1 (Smoothness)**.** The function f is continuously differentiable on $\mathbb{R}^d$, and its gradient ∇f is L_1-Lipschitz continuous, with $L_1 > 0$.

Assumption 2 (Lower bound)**.** There exists $f_{\text{low}} \in \mathbb{R}$ such that $f(x) \geq f_{\text{low}}$ for all $x \in \mathbb{R}^d$.

We further assume the strict saddle property; the precise statement (in terms of parameters $\zeta, \beta, \gamma, \delta$) is recalled in Section 2.2. We denote $g_k := \nabla f(x_k)$.

2.2 Strict saddle property

Definition 1 ([2])**.** A differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is $(\zeta, \beta, \gamma, \delta)$-*strict saddle* with parameters $\zeta \in (0, 1]$, $\beta > 0$, $\gamma > 0$, $\delta > 0$ if, for every $x \in \mathbb{R}^n$, at least one of the following holds:

1. **(Large gradient)** $\|\nabla f(x)\| \geq \zeta$,
2. **(Negative curvature)** $\lambda_{\min}(\nabla^2 f(x)) \leq -\beta$,
3. **(Near a local minimizer)** there exists a local minimizer $x^\star$ with $\|x - x^\star\| \leq \delta$ such that f is γ-strongly convex on $B(x^\star, 2\delta)$.

Remark 1. The restriction $\zeta \leq 1$ is a normalization convention used in the complexity analysis: it ensures that for any exponent $a \geq 0$, ζ^a is non-increasing in a, which simplifies the form of the Regime 1 bound (Theorem 1).

2.3 Hybrid algorithm with line search

We analyze the hybrid algorithm presented in Algorithm 1. The algorithm interleaves an NCG step (when the gradient norm exceeds the tolerance ε_g) with a negative-curvature step (when the smallest eigenvalue of the Hessian is below $-\varepsilon_H$). The NCG step uses backtracking Armijo line search initialized at $\alpha = 1$, with contraction factor $\theta \in (0, 1)$ and Armijo constant $\eta \in (0, 1)$. The conjugate direction is reset to the negative gradient when a modified restart condition is triggered.

Algorithm 1 Hybrid NCG / Negative-Curvature Algorithm with Line Search

Require: $x_0 \in \mathbb{R}^d$, $\eta \in (0,1)$, $\theta \in (0,1)$, $\sigma \in (0,1]$, $\kappa \geq 1$, $p \geq 0$, $q \geq 0$, $\varepsilon_g > 0$, $\varepsilon_H > 0$.

Set $g_0 = \nabla f(x_0)$, $d_0 = -g_0$, $k = 0$.
for $k = 0, 1, 2, \ldots$ **do**
  Compute $g_k = \nabla f(x_k)$.
  if $\|g_k\| \geq \varepsilon_g$ **then** ▷ NCG phase
    Compute $\alpha_k = \theta^{j_k}$, where j_k is the smallest nonnegative integer such that

$$f(x_k + \alpha_k d_k) \leq f(x_k) + \eta \, \alpha_k \, g_k^\top d_k. \tag{2}$$

    Set $x_{k+1} = x_k + \alpha_k d_k$ and $g_{k+1} = \nabla f(x_{k+1})$.
    Choose a conjugate direction parameter β_{k+1}.
    Set $d_{k+1} = -g_{k+1} + \beta_{k+1} d_k$.
    if the restart condition holds:

$$g_{k+1}^\top d_{k+1} \geq -\sigma \, \|g_{k+1}\|^{1+p} \quad \text{or} \quad \|d_{k+1}\| \geq \kappa \, \|g_{k+1}\|^q, \tag{3}$$

  then
      Restart: $d_{k+1} = -g_{k+1}$.
    end if
  else ▷ Negative-curvature phase
    Compute the smallest eigenvalue λ_k and an associated eigenvector e_k of $\nabla^2 f(x_k)$, with sign chosen so that $\langle g_k, e_k \rangle \leq 0$.
    if $-\lambda_k \geq \varepsilon_H$ **then**
      Set $x_{k+1} = x_k + \frac{2\lambda_k}{L_2} e_k$.
      Reset $d_{k+1} = -\nabla f(x_{k+1})$.
    else
      stop and **return** $x^\star = x_k$.
    end if
  end if
end for

2.4 Index partition

Following the convention of [1], we partition the NCG iterations according to whether the restart condition is triggered. Define

$$\mathcal{N} := \left\{k \in \mathbb{N} \;:\; g_k^\top d_k \le -\sigma \, \|g_k\|^{1+p} \quad \text{and} \quad \|d_k\| \le \kappa \, \|g_k\|^q \right\}, \tag{4}$$

$$\mathcal{R} := \mathbb{N} \setminus \mathcal{N}. \tag{5}$$

Indices in $\mathcal{N}$ correspond to non-restart NCG steps; indices in $\mathcal{R}$ correspond to restart steps (where $d_k = -g_k$).

3 Decrease Lemmas

We reuse the decrease lemmas of [1]; we recall them here for completeness, in the form needed in the sequel.

Lemma 1 ([1, Lemma 3.1])**.** *Let Assumption 1 hold, and let $k \in \mathcal{N}$ such that $\|g_k\| > 0$. Then, the line-search process terminates after at most $\lfloor \bar{\jmath}_{\mathcal{N},k} + 1 \rfloor$ iterations, where*

$$\bar{\jmath}_{\mathcal{N},k} := \left[\log_\theta \left(\frac{2(1-\eta)\,\sigma}{\kappa^2 \, L_1} \right) \|g_k\|^{1+p-2q} \right]_+ . \tag{6}$$

Moreover, the resulting decrease at the kth iteration satisfies

$$f(x_k) - f(x_{k+1}) \;>\; c_{\mathcal{N}} \, \min\left\{ \|g_k\|^{1+p}, \;\; \|g_k\|^{2(1+p-q)} \right\}, \tag{7}$$

where

$$c_{\mathcal{N}} := \eta\, \sigma \, \min\left\{ 1, \;\; \frac{2(1-\eta)\,\sigma\,\theta}{\kappa^2 \, L_1} \right\}. \tag{8}$$

Proof. See the proof of Lemma 3.1 in [1]. □

Lemma 2 ([1, Lemma 3.2])**.** *Let Assumption 1 hold, and let $k \in \mathcal{R}$ such that $\|g_k\| > 0$. Then, the line-search process terminates after at most $\bar{\jmath}_{\mathcal{R}} + 1$ iterations, where*

$$\bar{\jmath}_{\mathcal{R}} := \left[\log_\theta \left(\frac{2(1-\eta)}{L_1} \right) \right]_+ . \tag{9}$$

Moreover, the resulting decrease at the kth iteration satisfies

$$f(x_k) - f(x_{k+1}) \;>\; c_{\mathcal{R}} \, \|g_k\|^2, \tag{10}$$

where

$$c_{\mathcal{R}} := \eta \, \min\left\{ 1, \;\; \frac{2(1-\eta)\,\theta}{L_1} \right\}. \tag{11}$$

Proof. See the proof of Lemma 3.2 in [1]. □

Lemma 3 ([3, Theorem 9.5])**.** *Let A1–A2 hold and $-\lambda_k \ge \varepsilon_H > 0$. Then the step $x_{k+1} = x_k + \frac{2\lambda_k}{L_2} e_k$ satisfies:*

$$f(x_k) - f(x_{k+1}) \ge \frac{2|\lambda_k|^3}{3L_2^2} \ge \frac{2\varepsilon_H^3}{3L_2^2}. \tag{12}$$

Proof. See the proof of Theorem 9.5 in [3]. □

Remark 2. All three lemmas guarantee $f(x_{k+1}) < f(x_k)$. This monotone decrease property ensures iterates remain within sublevel sets of f.

4 Complexity in Regime 1: Large Gradient

Theorem 1 (Regime 1 complexity; adapted from [1, Theorem 3.1])**.** *Let Assumption 1 and Assumption 2 hold, and suppose that $1 + p - q \geq 0$. Then, the number of iterations spent in Regime 1 (i.e., iterations k with $\|g_k\| \geq \zeta$) by Algorithm 1 is at most*

$$K_1 \;\leq\; \left\lceil \frac{f(x_0) - f_{\text{low}}}{c_{\mathcal{R}}}\,\zeta^{-2} \;+\; \frac{f(x_0) - f_{\text{low}}}{c_{\mathcal{N}}}\,\zeta^{-\max\{1+p,\,2(1+p-q)\}} \right\rceil. \tag{13}$$

Proof. Let $K \in \mathbb{N}$ be such that $\|g_k\| \geq \zeta$ for every $k \in \{0, \ldots, K-1\}$. Following the partition (4)–(5), define

$$\mathcal{N}_K := \mathcal{N} \cap \{0, \ldots, K-1\},$$
$$\mathcal{R}_K := \mathcal{R} \cap \{0, \ldots, K-1\}.$$

For any $k \in \mathcal{N}_K$, Lemma 1 applies, and

$$f(x_k) - f(x_{k+1}) \;\geq\; c_{\mathcal{N}}\, \min\Big\{ \|g_k\|^{1+p},\;\; \|g_k\|^{2(1+p-q)} \Big\} \;\geq\; c_{\mathcal{N}}\, \zeta^{\max\{1+p,\,2(1+p-q)\}}, \tag{14}$$

where the last inequality uses $\|g_k\| \geq \zeta$ together with $1 + p - q \geq 0$ (so that both exponents are nonnegative and the worst case for $\|g_k\|^r \geq \zeta^r$ corresponds to the larger exponent).

For any $k \in \mathcal{R}_K$, Lemma 2 applies, and

$$f(x_k) - f(x_{k+1}) \;\geq\; c_{\mathcal{R}}\, \|g_k\|^2 \;\geq\; c_{\mathcal{R}}\, \zeta^2. \tag{15}$$

Summing over $k \in \{0, \ldots, K-1\}$ and using Assumption 2,

$$\begin{aligned} f(x_0) - f_{\text{low}} \;&\geq\; f(x_0) - f(x_K) \;=\; \sum_{k=0}^{K-1} \big[f(x_k) - f(x_{k+1}) \big] \\ &=\; \sum_{k \in \mathcal{N}_K} \big[f(x_k) - f(x_{k+1}) \big] + \sum_{k \in \mathcal{R}_K} \big[f(x_k) - f(x_{k+1}) \big] \\ &\geq\; |\mathcal{N}_K|\, c_{\mathcal{N}}\, \zeta^{\max\{1+p,\,2(1+p-q)\}} + |\mathcal{R}_K|\, c_{\mathcal{R}}\, \zeta^2. \end{aligned}$$

Since both terms on the right-hand side are nonnegative, the inequality implies

$$|\mathcal{N}_K| \;<\; \frac{f(x_0) - f_{\text{low}}}{c_{\mathcal{N}}}\, \zeta^{-\max\{1+p,\,2(1+p-q)\}} \quad \text{and} \quad |\mathcal{R}_K| \;<\; \frac{f(x_0) - f_{\text{low}}}{c_{\mathcal{R}}}\, \zeta^{-2}.$$

Using $|\mathcal{N}_K| + |\mathcal{R}_K| = K$ yields

$$K \;<\; \frac{f(x_0) - f_{\text{low}}}{c_{\mathcal{R}}}\, \zeta^{-2} + \frac{f(x_0) - f_{\text{low}}}{c_{\mathcal{N}}}\, \zeta^{-\max\{1+p,\,2(1+p-q)\}},$$

which establishes (13). ◻

5 Complexity in Regime 2: Negative Curvature

Theorem 2 (Regime 2 complexity)**.** *Let Assumption 1 and Assumption 2 hold, and suppose that f satisfies the $(\zeta, \beta, \gamma, \delta)$-strict saddle property (Definition 1). Then the number of iterations spent in Regime 2 by Algorithm 1 is at most*

$$K_2 \;\leq\; \left\lceil \frac{3\, L_2^2\, (f(x_0) - f_{\text{low}})}{2\, \beta^3} \right\rceil. \tag{16}$$

Proof. At any iteration k in Regime 2, the iterate x_k satisfies $\|g_k\| < \zeta \le \varepsilon_g$ and $\lambda_{\min}\big(\nabla^2 f(x_k)\big) \le -\beta$. Algorithm 1 therefore enters the negative-curvature branch, and since $-\lambda_k \ge \beta \ge \varepsilon_H$ (assuming $\varepsilon_H \le \beta$), the negative-curvature step is taken. By Lemma 3,

$$f(x_k) - f(x_{k+1}) \;\ge\; \frac{2\,|\lambda_k|^3}{3\,L_2^2} \;\ge\; \frac{2\,\beta^3}{3\,L_2^2}.$$

Summing over the K_2 Regime 2 iterations and using Assumption 2,

$$f(x_0) - f_{\text{low}} \;\ge\; f(x_0) - f(x_{K_2}) \;\ge\; K_2 \cdot \frac{2\,\beta^3}{3\,L_2^2},$$

which gives (16). □

6 Complexity in Regime 3: Strong Convexity

6.1 Setting

Let $x^\star$ denote the local minimizer associated with the Regime 3 ball: by the strict saddle property (Definition 1), there exists $x^\star$ with $\|x_{k_0} - x^\star\| \le \delta$ such that f is γ-strongly convex on $\bar{B}(x^\star, 2\delta)$, where k_0 denotes the first iteration entering Regime 3. We write

$$\Delta_k \;:=\; f(x_k) - f(x^\star) \tag{17}$$

for the suboptimality at iterate k, and we adopt the shorthand

$$a := 1 + p, \qquad b := 2(1 + p - q), \qquad R := \frac{\max(a, b)}{2}. \tag{18}$$

Standing inequalities on $\bar{B}(x^\star, 2\delta)$. For any $x_k \in \bar{B}(x^\star, 2\delta)$, smoothness and strong convexity give, respectively,

$$\Delta_k \;\le\; \tfrac{L_1}{2}\,\|x_k - x^\star\|^2, \tag{19}$$

$$\Delta_k \;\ge\; \tfrac{\gamma}{2}\,\|x_k - x^\star\|^2, \tag{20}$$

$$2\gamma\,\Delta_k \;\le\; \|g_k\|^2 \;\le\; 2L_1\,\Delta_k, \tag{21}$$

where (21) follows from (19)–(20) and the Polyak–Łojasiewicz inequality.

Entry value and gradient bound. At the entry iterate k_0, $\|x_{k_0} - x^\star\| \le \delta$ and (19) give

$$\Delta_{k_0} \;\le\; \frac{L_1\,\delta^2}{2}. \tag{22}$$

Combined with monotonicity of Δ_k (which holds along the line search by Armijo) and (21),

$$\|g_k\| \;\le\; L_1\,\delta \qquad \text{for all } k \ge k_0 \text{ with } x_k \in \bar{B}(x^\star, 2\delta). \tag{23}$$

Stopping criterion. The algorithm exits Regime 3 once $\|g_k\| \le \varepsilon_g$. By (21), a sufficient condition is

$$\Delta_k \;\le\; \frac{\varepsilon_g^2}{2L_1}, \tag{24}$$

which is the form we use to count iterations.

Per-iteration decrease. We split Regime 3 iterations according to whether the restart condition is triggered. For non-restart steps, $k \in \mathcal{N}$, Lemma 1 gives

$$\Delta_k - \Delta_{k+1} \;\geq\; c_{\mathcal{N}} \min\Big\{ \|g_k\|^a \,,\; \|g_k\|^b \Big\}\,, \tag{25}$$

where $a = 1+p$ and $b = 2(1+p-q)$. For restart steps, $k \in \mathcal{R}$, Lemma 2 gives

$$\Delta_k - \Delta_{k+1} \;\geq\; c_{\mathcal{R}}\, \|g_k\|^2 \;\geq\; 2\gamma\, c_{\mathcal{R}}\, \Delta_k, \tag{26}$$

where the second inequality uses the strong-convexity bound (21).

The restart-step recurrence (26) is a clean linear contraction and yields the bound

$$K_3^{\mathcal{R}} \;\leq\; \left\lceil \frac{1}{2\gamma\, c_{\mathcal{R}}} \log\!\left(\frac{L_1^2\, \delta^2}{\varepsilon_g^2} \right) \right\rceil \tag{27}$$

on the number of restart steps in Regime 3, by the same argument as in Subcase L1 below.

For non-restart steps, the presence of the minimum in (25) forces a phase split based on the magnitude of $\|g_k\|$, and subsequently a subcase analysis based on the value of $R := \max(a,b)/2$.

6.2 Phase split

Since $t \mapsto t^s$ is increasing for $t \geq 1$ and decreasing for $0 < t \leq 1$,

$$\min\big(\|g_k\|^a \,,\; \|g_k\|^b\big) \;=\; \begin{cases} \|g_k\|^{\min(a,b)} & \text{if } \|g_k\| \geq 1, \\ \|g_k\|^{\max(a,b)} = \|g_k\|^{2R} & \text{if } \|g_k\| \leq 1. \end{cases} \tag{28}$$

Accordingly, we split the Regime 3 iterations into two phases:

- *Phase H* (high gradient): iterations k with $\|g_k\| \geq 1$.
- *Phase L* (low gradient): iterations k with $\|g_k\| \leq 1$.

We denote by K_3^H and K_3^L the respective iteration counts; the total Regime 3 count is $K_3 = K_3^H + K_3^L$.

6.3 Phase H bound

We split Phase H by the sign of $b = 2(1+p-q)$, since $a = 1+p \geq 1 > 0$ always.

Lemma 4 (Phase H bound)**.** *Suppose $x_k \in \bar{B}(x^\star, 2\delta)$ for every iterate k in Phase H. Then:*

(i) If $q \leq 1+p$ (so $b \geq 0$), then

$$K_3^H \;\leq\; \left\lceil \frac{L_1\, \delta^2}{2\, c_{\mathcal{N}}} \right\rceil. \tag{29}$$

(ii) If $q > 1+p$ (so $b < 0$), then

$$K_3^H \;\leq\; \left\lceil \frac{L_1\, \delta^2\, (L_1\, \delta)^{2(q-1-p)}}{2\, c_{\mathcal{N}}} \right\rceil. \tag{30}$$

Proof. By (25) and the fact that $\|g_k\| \geq 1$ in Phase H,

$$\Delta_k - \Delta_{k+1} \;\geq\; c_{\mathcal{N}}\, \|g_k\|^{\min(a,b)}\,.$$

Case (i): $q \le 1+p$**.** Then $b \ge 0$, hence $\min(a,b) \ge 0$ and $\|g_k\|^{\min(a,b)} \ge 1$. Summing over Phase H iterations and using monotonicity of Δ_k together with (22),

$$\frac{L_1\,\delta^2}{2} \;\ge\; \Delta_{k_0} \;\ge\; \sum_{k\in\text{Phase H}} (\Delta_k - \Delta_{k+1}) \;\ge\; K_3^H \cdot c_{\mathcal{N}},$$

which yields (29).

Case (ii): $q > 1+p$**.** Then $b < 0$ and $\min(a,b) = b$. The quantity $\|g_k\|^b$ is small when $\|g_k\|$ is large, so we apply the upper bound $\|g_k\| \le L_1\delta$ from (23):

$$\|g_k\|^b \;\ge\; (L_1\delta)^b \;=\; (L_1\delta)^{-2(q-1-p)}.$$

Hence $\Delta_k - \Delta_{k+1} \ge c_{\mathcal{N}}\,(L_1\delta)^{-2(q-1-p)}$. Summing as in Case (i):

$$\frac{L_1\,\delta^2}{2} \;\ge\; K_3^H \cdot c_{\mathcal{N}}\,(L_1\delta)^{-2(q-1-p)},$$

which yields (30). □

Remark 3. Case (ii) is significantly worse than Case (i): the bound depends on a positive power of $L_1\delta$, which can be large. In what follows we focus on the favorable regime $q \le 1+p$.

6.4 Phase L analysis

For k in Phase L (i.e., $\|g_k\| \le 1$), the per-iteration decrease (25) together with $\max(a,b) = 2R$ and the strong-convexity inequality (21) yields

$$\Delta_k - \Delta_{k+1} \;\ge\; c_{\mathcal{N}}\,\|g_k\|^{\max(a,b)} \;=\; c_{\mathcal{N}}\,(\|g_k\|^2)^R \;\ge\; c_{\mathcal{N}}\,(2\gamma\,\Delta_k)^R.$$

Equivalently,

$$\Delta_{k+1} \;\le\; \bigl(1 - c_{\mathcal{N}}\,(2\gamma)^R\,\Delta_k^{R-1}\bigr)\,\Delta_k. \tag{31}$$

The behavior of K_3^L splits into three subcases according to the value of R. Throughout this subsection we let k_1 denote the first iteration of Phase L; by monotonicity $\Delta_{k_1} \le \Delta_{k_0} \le L_1\delta^2/2$.

Subcase L1: $R = 1$ **(linear convergence).** This subcase corresponds to $p = 1$ and $q \ge 1$, or $p \le 1$ and $q = p$.

Lemma 5 (Subcase L1)**.** *Suppose $x_k \in \bar{B}(x^\star, 2\delta)$ for every iterate k in Phase L, $R = 1$, and $2\,c_{\mathcal{N}}\,\gamma < 1$. Then*

$$K_3^L \;\le\; \left\lceil \frac{1}{2\,c_{\mathcal{N}}\,\gamma}\,\log\!\left(\frac{L_1^2\,\delta^2}{\varepsilon_g^2}\right)\right\rceil. \tag{32}$$

Proof. With $R = 1$, the recurrence (31) simplifies to $\Delta_{k+1} \le (1 - 2\,c_{\mathcal{N}}\,\gamma)\,\Delta_k$. Iterating from k_1,

$$\Delta_{k_1+K} \;\le\; (1 - 2\,c_{\mathcal{N}}\,\gamma)^K\,\Delta_{k_1}.$$

A sufficient condition for $\Delta_{k_1+K} \le \varepsilon_g^2/(2L_1)$ is $(1-2\,c_{\mathcal{N}}\,\gamma)^K\,\Delta_{k_1} \le \varepsilon_g^2/(2L_1)$. Taking logarithms and dividing by $\log(1 - 2\,c_{\mathcal{N}}\,\gamma) < 0$ (which flips the inequality):

$$K \;\ge\; \frac{\log\bigl(2L_1\Delta_{k_1}/\varepsilon_g^2\bigr)}{-\log(1 - 2\,c_{\mathcal{N}}\,\gamma)} \;\ge\; \frac{\log\bigl(2L_1\Delta_{k_1}/\varepsilon_g^2\bigr)}{2\,c_{\mathcal{N}}\,\gamma},$$

where the last inequality uses $-\log(1-x) \ge x$ for $x \in [0,1)$. Substituting $\Delta_{k_1} \le L_1\delta^2/2$ gives $2L_1\Delta_{k_1}/\varepsilon_g^2 \le L_1^2\delta^2/\varepsilon_g^2$, which yields (32). □

Subcase L2: $R < 1$ (finite termination). This subcase corresponds to $p < 1$ and $q > p$.

Lemma 6 (Subcase L2)**.** *Suppose $x_k \in \bar{B}(x^\star, 2\delta)$ for every iterate k in Phase L, and $R < 1$. Then*

$$K_3^L \;\leq\; \left\lceil \frac{(L_1\delta^2/2)^{1-R}}{(1-R)\, c_{\mathcal{N}}\, (2\gamma)^R} \right\rceil. \tag{33}$$

Proof. Define the Lyapunov function $\varphi_k := \Delta_k^{1-R}$. Since $1 - R \in (0, 1)$, the map $t \mapsto t^{1-R}$ is increasing and concave on $(0, \infty)$. Raising both sides of (31) to the power $1 - R$ (monotone increasing):

$$\varphi_{k+1} \;\leq\; \big(1 - c_{\mathcal{N}}(2\gamma)^R \Delta_k^{R-1}\big)^{1-R}\, \varphi_k.$$

Concavity gives $(1 - c)^{1-R} \leq 1 - (1 - R)c$ for $c \in (0, 1)$, so

$$\varphi_{k+1} \;\leq\; \big(1 - (1-R)c_{\mathcal{N}}(2\gamma)^R \Delta_k^{R-1}\big)\, \varphi_k \;=\; \varphi_k - (1-R)c_{\mathcal{N}}(2\gamma)^R,$$

where the last equality uses $\Delta_k^{R-1} \cdot \varphi_k = \Delta_k^{R-1} \cdot \Delta_k^{1-R} = 1$. Telescoping from k_1:

$$\varphi_{k_1+K} \;\leq\; \varphi_{k_1} - (1-R)c_{\mathcal{N}}(2\gamma)^R\, K.$$

Since $\varphi_k \geq 0$ for all k,

$$K_3^L \;\leq\; \frac{\varphi_{k_1}}{(1-R)\, c_{\mathcal{N}}\, (2\gamma)^R} \;\leq\; \frac{(L_1\delta^2/2)^{1-R}}{(1-R)\, c_{\mathcal{N}}\, (2\gamma)^R},$$

where we used $\varphi_{k_1} = \Delta_{k_1}^{1-R} \leq (L_1\delta^2/2)^{1-R}$. □

Subcase L3: $R > 1$ (sublinear convergence). This subcase corresponds to $p > 1$ or $q < p$.

Lemma 7 (Subcase L3)**.** *Suppose $x_k \in \bar{B}(x^\star, 2\delta)$ for every iterate k in Phase L, and $R > 1$. Suppose further that $c_{\mathcal{N}}(2\gamma)^R \Delta_{k_1}^{R-1} < 1$. Then*

$$K_3^L \;\leq\; \left\lceil \frac{(2L_1)^{R-1}}{(R-1)\, c_{\mathcal{N}}\, (2\gamma)^R\, \varepsilon_g^{2(R-1)}} \right\rceil. \tag{34}$$

Proof. Define $\varphi_k := \Delta_k^{1-R}$. Since $1 - R < 0$, the map $t \mapsto t^{1-R}$ is decreasing and convex on $(0, \infty)$. Raising both sides of (31) to the power $1 - R$ (decreasing, so the inequality flips):

$$\varphi_{k+1} \;\geq\; \big(1 - c_{\mathcal{N}}(2\gamma)^R \Delta_k^{R-1}\big)^{1-R}\, \varphi_k.$$

Convexity gives $(1 - c)^{1-R} \geq 1 - (1 - R)c$ for $c \in (0, 1)$, so

$$\varphi_{k+1} \;\geq\; \big(1 - (1-R)c_{\mathcal{N}}(2\gamma)^R \Delta_k^{R-1}\big)\, \varphi_k \;=\; \varphi_k + (R-1)c_{\mathcal{N}}(2\gamma)^R.$$

Telescoping:

$$\varphi_{k_1+K} \;\geq\; \varphi_{k_1} + (R-1)c_{\mathcal{N}}(2\gamma)^R\, K.$$

The stopping criterion $\Delta_k \leq \varepsilon_g^2/(2L_1)$ is equivalent to $\varphi_k \geq (2L_1/\varepsilon_g^2)^{R-1}$ (raising to the negative power $1 - R$ flips the inequality). A sufficient condition for termination by iteration $k_1 + K$ is therefore

$$\varphi_{k_1} + (R-1)c_{\mathcal{N}}(2\gamma)^R\, K \;\geq\; (2L_1/\varepsilon_g^2)^{R-1},$$

which gives

$$K \;\geq\; \frac{(2L_1/\varepsilon_g^2)^{R-1} - \varphi_{k_1}}{(R-1)\, c_{\mathcal{N}}\, (2\gamma)^R}.$$

Since $\varphi_{k_1} \geq 0$, dropping $-\varphi_{k_1}$ yields the looser but explicit upper bound (34). □

Remark 4. Unlike Subcases L1 and L2, the bound (34) does not depend on δ. In the sublinear regime, the dominant cost is reaching the small target tolerance ε_g, not handling the entry value Δ_{k_1}.

6.5 Iterate confinement

[Staying in strong convexity region, and / or finding the $m^\star$.]

6.6 Total Regime 3 bound

The total Regime 3 iteration count splits into restart and non-restart contributions:

$$K_3 \;=\; K_3^{\mathcal{R}} + K_3^{\mathcal{N}}, \qquad K_3^{\mathcal{N}} = K_3^{H} + K_3^{L},$$

where $K_3^{\mathcal{R}}$ counts restart steps and $K_3^{\mathcal{N}} = K_3^{H} + K_3^{L}$ counts non-restart steps split by the Phase H / Phase L analysis.

Lemma 8 (Restart-step bound)**.** *Suppose $x_k \in \bar{B}(x^\star, 2\delta)$ for every restart step in Regime 3, and $2\gamma\, c_{\mathcal{R}} < 1$. Then*

$$K_3^{\mathcal{R}} \;\le\; \left\lceil \frac{1}{2\,\gamma\, c_{\mathcal{R}}} \log\!\left(\frac{L_1^2\,\delta^2}{\varepsilon_g^2}\right)\right\rceil. \tag{35}$$

Proof. By (26), the restart-step recurrence is $\Delta_{k+1} \le (1 - 2\gamma\, c_{\mathcal{R}})\,\Delta_k$. The bound follows by the same argument as in the proof of Lemma 5, with $c_{\mathcal{N}}\,\gamma$ replaced by $\gamma\, c_{\mathcal{R}}$ and Δ_{k_1} replaced by the Regime 3 entry value $\Delta_{k_0} \le L_1\delta^2/2$. ☐

Theorem 3 (Regime 3 complexity)**.** *Suppose f satisfies the strict saddle property (Definition 1) and that $x_k \in \bar{B}(x^\star, 2\delta)$ for every Regime 3 iterate. Suppose moreover that $q \le 1+p$ and (where applicable) the smallness conditions of Lemma 8, Lemma 5, and Lemma 7 hold. Then the number of iterations spent in Regime 3 by Algorithm 1 satisfies*

$$K_3 \;\le\; \underbrace{\left\lceil \frac{1}{2\,\gamma\, c_{\mathcal{R}}} \log\!\left(\frac{L_1^2\,\delta^2}{\varepsilon_g^2}\right)\right\rceil}_{K_3^{\mathcal{R}}} + \underbrace{\left\lceil \frac{L_1\,\delta^2}{2\,c_{\mathcal{N}}}\right\rceil}_{K_3^{H}} + \underbrace{\begin{cases} \left\lceil \dfrac{1}{2\,c_{\mathcal{N}}\,\gamma} \log\!\left(\dfrac{L_1^2\,\delta^2}{\varepsilon_g^2}\right)\right\rceil & \text{if } R = 1,\\[2ex] \left\lceil \dfrac{(L_1\delta^2/2)^{1-R}}{(1-R)\,c_{\mathcal{N}}\,(2\gamma)^R}\right\rceil & \text{if } R < 1,\\[2ex] \left\lceil \dfrac{(2L_1)^{R-1}}{(R-1)\,c_{\mathcal{N}}\,(2\gamma)^R\,\varepsilon_g^{2(R-1)}}\right\rceil & \text{if } R > 1,\end{cases}}_{K_3^{L}} \tag{36}$$

where $R := \max(1+p,\, 2(1+p-q))/2$.

Proof. Direct combination of Lemma 8 for restart steps, Lemma 4 (case (i), valid under $q \le 1+p$) for non-restart Phase H steps, and whichever of Lemma 5, Lemma 6, Lemma 7 applies according to the value of R, for non-restart Phase L steps. ☐

Corollary 1 (Asymptotic Regime 3 complexity)**.** *Under the hypotheses of Theorem 3, as $\varepsilon_g \to 0$ with the strict-saddle parameters $(\zeta, \beta, \gamma, \delta)$ held fixed,*

$$K_3 \;=\; \begin{cases} \mathcal{O}\big(\log(1/\varepsilon_g)\big) & \text{if } R = 1,\\ \mathcal{O}\big(\log(1/\varepsilon_g)\big) & \text{if } R < 1,\\ \mathcal{O}\big(\varepsilon_g^{-2(R-1)}\big) & \text{if } R > 1,\end{cases} \tag{37}$$

where $R := \max(1+p,\, 2(1+p-q))/2$. *The constants hide dependence on $L_1, \gamma, c_{\mathcal{R}}, c_{\mathcal{N}}, \delta, R$ but not on ε_g.*

Remark 5 (Restart steps dominate the $R < 1$ case)**.** For $R < 1$, the non-restart Phase L analysis predicts a constant number of iterations, $K_3^L = \mathcal{O}(1)$. However, restart steps in Regime 3 contribute $K_3^{\mathcal{R}} = \mathcal{O}(\log(1/\varepsilon_g))$ regardless of R. Thus the total Regime 3 cost is $\mathcal{O}(\log(1/\varepsilon_g))$, not $\mathcal{O}(1)$ as the non-restart analysis alone would suggest.

The question of how often restart steps are triggered in Regime 3, and in particular, whether the restart conditions (3) are satisfied infinitely often near a strict local minimizer is algorithm- and parameter-dependent and beyond the scope of this analysis.

7 Total Complexity

We now combine the per-regime bounds of Theorems 1 to 3 into a complexity bound on the total number of iterations performed by Algorithm 1.

7.1 Disjointness of regimes

At each iteration k before termination, the iterate x_k falls into exactly one of the three regimes of the strict saddle decomposition (Definition 1):

- *Regime 1*: $\|g_k\| \geq \zeta$. Algorithm 1 performs an NCG step with line search.
- *Regime 2*: $\|g_k\| < \zeta$ and $\lambda_{\min}(\nabla^2 f(x_k)) \leq -\beta$. Algorithm 1 performs a negative-curvature step.
- *Regime 3*: $\|g_k\| < \zeta$ and $\lambda_{\min}(\nabla^2 f(x_k)) > -\beta$. The iterate lies in the strong-convexity neighborhood of a local minimizer.

Denote by K_1, K_2, K_3 the number of iterations spent in each regime. The total iteration count of the algorithm is

$$K_{\text{total}} \;=\; K_1 + K_2 + K_3. \tag{38}$$

7.2 Total complexity bound

Theorem 4 (Total complexity)**.** *Let Assumption 1 and Assumption 2 hold, and let f satisfy the $(\zeta, \beta, \gamma, \delta)$-strict saddle property (Definition 1). Suppose:*

(i) the algorithm parameters satisfy $1 + p - q \geq 0$ and $q \leq 1 + p$;

(ii) the tolerances satisfy $\varepsilon_g \leq \zeta$ and $\varepsilon_H \leq \beta$;

(iii) the iterates remain in $\bar{B}(x^\star, 2\delta)$ throughout Regime 3;

(iv) the smallness conditions of Lemma 8, Lemma 5, and Lemma 7 hold where applicable.

Then the total number of iterations performed by Algorithm 1 satisfies

$$K_{\text{total}} \;\leq\; K_1 + K_2 + K_3, \tag{39}$$

where K_1, K_2, K_3 are bounded as in Theorem 1, Theorem 2, and Theorem 3, respectively.

Proof. By the partition above, every iteration before termination falls into exactly one regime. Summing the per-regime bounds of Theorems 1 to 3 gives (39). $\square$

7.3 Asymptotic behavior

The bound (39) is best understood asymptotically. The per-regime rates are:

- $K_1 = \mathcal{O}\big(\zeta^{-\max\{1+p,\,2(1+p-q)\}}\big) = \mathcal{O}\big(\zeta^{-2R}\big)$ (from Theorem 1, since $\zeta \leq 1$).
- $K_2 = \mathcal{O}\big(\beta^{-3}\big)$ (from Theorem 2).
- K_3 as in Corollary 1.

Under the strict-saddle framing, $\zeta, \beta, \gamma, \delta$ are properties of the function (fixed by the strict saddle decomposition), while ε_g is the user-chosen tolerance scaling toward 0. The asymptotic complexity in ε_g is therefore governed by K_3 alone, since K_1 and K_2 are independent of ε_g.

Corollary 2 (Asymptotic total complexity)**.** *Fix a function f satisfying the $(\zeta, \beta, \gamma, \delta)$-strict saddle property, so that $\zeta, \beta, \gamma, \delta$ are absolute constants of the problem. Under the hypotheses of Theorem 4, as $\varepsilon_g \to 0$ with $\zeta, \beta, \gamma, \delta$ held fixed,*

$$K_{\text{total}} = \begin{cases} \mathcal{O}\big(\log(1/\varepsilon_g)\big) & \text{if } R \leq 1, \\ \mathcal{O}\big(\varepsilon_g^{-2(R-1)}\big) & \text{if } R > 1, \end{cases} \tag{40}$$

where $R := \max(1+p,\, 2(1+p-q))/2$. The constants hide dependence on $(\zeta, \beta, \gamma, \delta)$ and on the algorithm and Lipschitz parameters, but not on ε_g.

Remark 6 (The role of restart steps in Regime 3)**.** The case $R < 1$ deserves particular attention. The non-restart Phase L analysis predicts a constant number of iterations, which would suggest $K_3 = \mathcal{O}(1)$ and a total complexity dominated by $K_1 + K_2$ (independent of ε_g). However, restart steps in Regime 3 contribute $K_3^{\mathcal{R}} = \mathcal{O}(\log(1/\varepsilon_g))$ regardless of R, and as long as restart steps are triggered in Regime 3, they set a logarithmic floor on the total complexity. This is reflected in (40) by merging the $R = 1$ and $R < 1$ cases into a single $\mathcal{O}(\log(1/\varepsilon_g))$ rate.

Remark 7 (The role of Regime 3)**.** Corollary 2 reveals the qualitative impact of the strong-convexity phase: although the algorithm performs work in all three regimes, only Regime 3 produces an ε_g-dependent contribution. In the favorable parameter regime ($R \leq 1$), this contribution is logarithmic, so the algorithm achieves an asymptotic complexity of $\mathcal{O}(\log(1/\varepsilon_g))$ on top of the saddle-escape cost $K_1 + K_2$. In the unfavorable regime ($R > 1$), the local-minimizer phase becomes the dominant cost, with sublinear rate $\mathcal{O}(\varepsilon_g^{-2(R-1)})$.

Remark 8 (Two distinct asymptotic regimes)**.** Corollary 2 treats $\zeta, \beta, \gamma, \delta$ as fixed properties of the function f and lets only the user-chosen tolerance ε_g scale toward 0. This is the natural regime under the strict saddle assumption, where the function-class parameters are intrinsic to f.

A second possible regime couples ζ to ε_g and β to ε_H (e.g., $\zeta = \varepsilon_g$ and $\beta = \varepsilon_H$). In that regime, K_1 and K_2 contribute ε_g- and ε_H-dependent terms ($\mathcal{O}(\varepsilon_g^{-2R})$ and $\mathcal{O}(\varepsilon_H^{-3})$ respectively), and the dominant term is the worst of the three. Our framing is the strict-saddle framing; the second regime is not the subject of this paper.

8 Discussion

[Comparison with fixed-stepsize results, limitations of proof techniques ?]

A Auxiliary results

[If needed: technical lemmas, etc]

References

[1] Rémi Chan–Renous-Legoubin and Clément W. Royer. A nonlinear conjugate gradient method with complexity guarantees and its application to nonconvex regression. *EURO Journal on Computational Optimization*, 10:100044, 2022.

[2] Rong Ge, Furong Huang, Chi Jin, and Yang Yuan. Escaping from saddle points — online stochastic gradient for tensor decomposition. In Peter Grünwald, Elad Hazan, and Satyen Kale, editors, *Proceedings of The 28th Conference on Learning Theory*, volume 40 of *Proceedings of Machine Learning Research*, pages 797–842, Paris, France, 03–06 Jul 2015. PMLR.

[3] John Wright and Yi Ma. *High-Dimensional Data Analysis with Low-Dimensional Models: Principles, Computation, and Applications*. Cambridge University Press, 2022.